



1 GENERAL INFORMATION

DESIGNERS

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INSTITUTION

Institute of Materials Research and Engineering (IMRE),
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2 DESCRIPTION

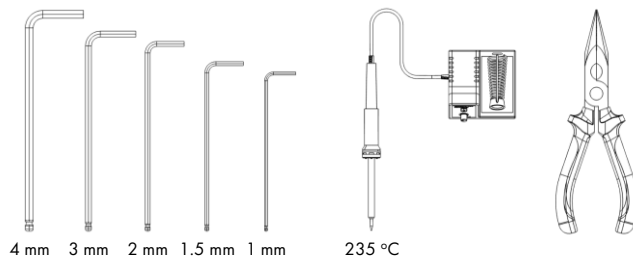
This is the assembly guide for the Qubot Gantry module.
Version: V4
*Refer to separate manual for microcontroller and wiring.
*Utility frame not included

3 SKILLS & RESOURCES

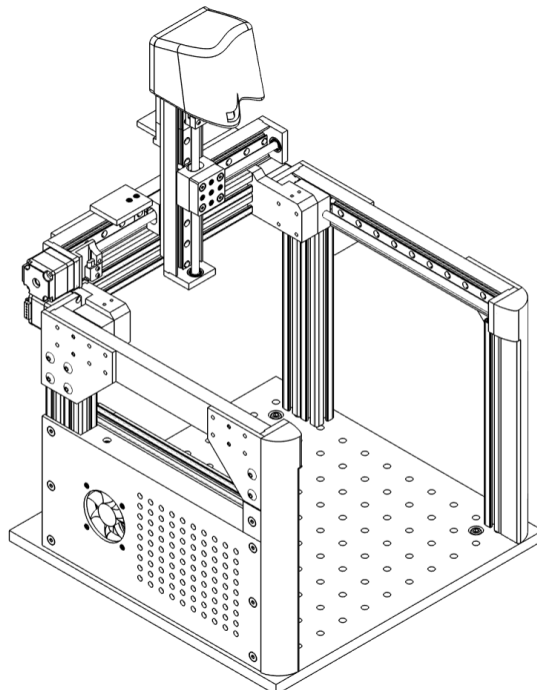
Access to FDM 3D printing
Access to metal CNC milling
Access to acrylic laser cutting
Heat insert

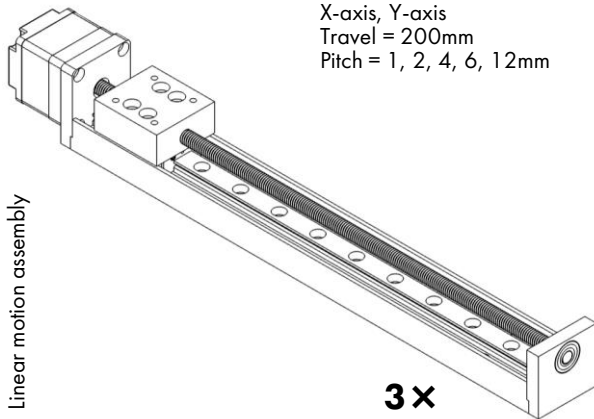
Required
Optional
Optional
Required

4 TOOLS



5 MODEL

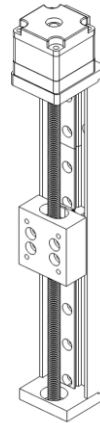




X-axis, Y-axis
Travel = 200mm
Pitch = 1, 2, 4, 6, 12mm

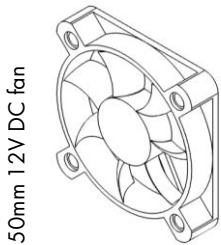
Linear motion assembly

3x



Z-axis
Travel = 50, 100, 150, 200mm
Pitch = 1, 2, 4mm

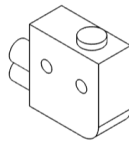
1x



50mm 12V DC fan

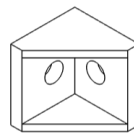
1x

Endstop switch



3x

2020 Bracket



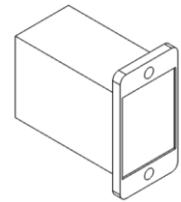
2x

Rubber foot M6

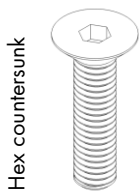


3x

Panel mount USB



1x



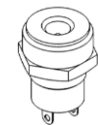
Hex countersunk

M3x06 **2x**
M3x10 **4x**
M3x14 **4x**

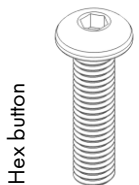
M4x08 **12x**
M4x10 **2x**
M4x14 **4x**

M5x10 **4x**
M6x10 **4x**

Panel mount DC jack



1x



Hex button

M2x08 **6x**
M2.5x06 **2x**
M2.5x14 **4x**

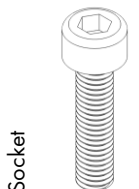
M3x08 **5x**
M3x12 **4x**

M5x08 **22x**
M6x10 **2x**

Drop-in T-nut 2020



M3 **1x**
M4 **16x**
M5 **22x**



Socket

M3x06 **12x**
M3x12 **4x**
M3x25 **8x**
M5x20 **6x**

Knurled heat insert



M2xL3xOD3.2 **6x**
M2.5xL3xOD3.5 **2x**
M3xL3xOD4.5 **21x**

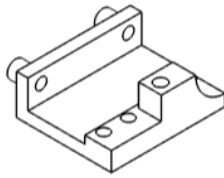
Dome nut



M3 **1x**

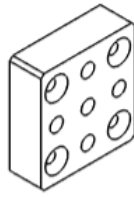


X and Z endstop mount



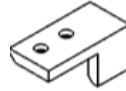
2x ↑

Z-axis mount



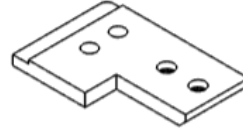
1x ↗

Y endstop trigger



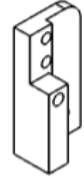
1x ↗

Y cable chain mount



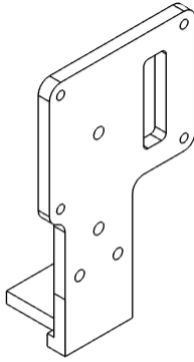
1x ↑

Y endstop mount



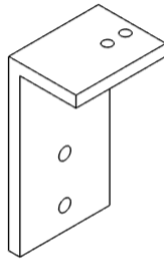
1x ↗

Z platform



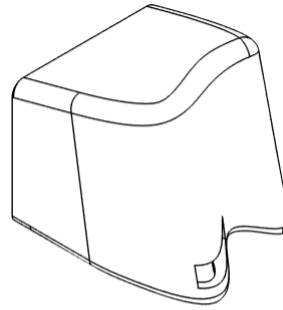
1x ↗

Z cable chain mount



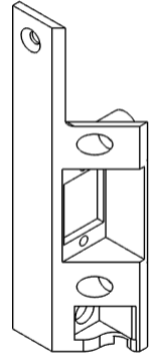
1x ↗

Z casing



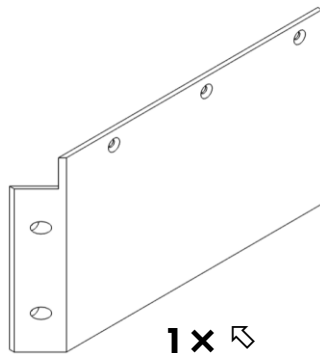
1x ↓

Electronic casing back



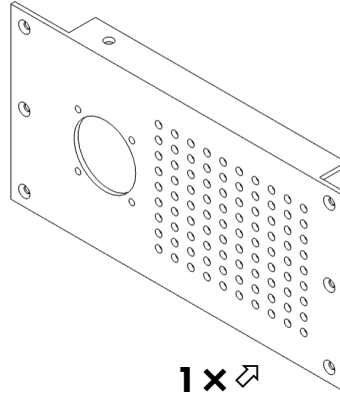
1x ↗

Electronic casing front



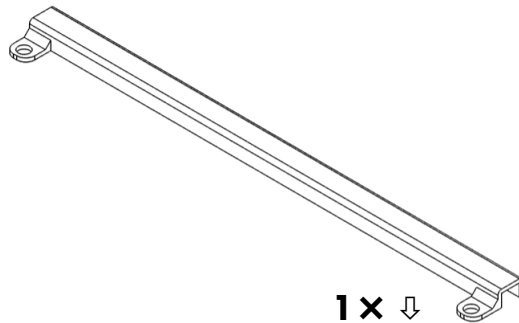
1x ↗

Electronic casing base



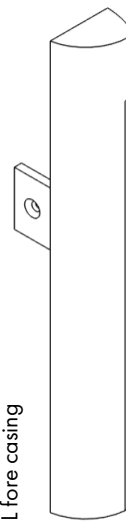
1x ↗

Cable track



1x ↓

L fore casing

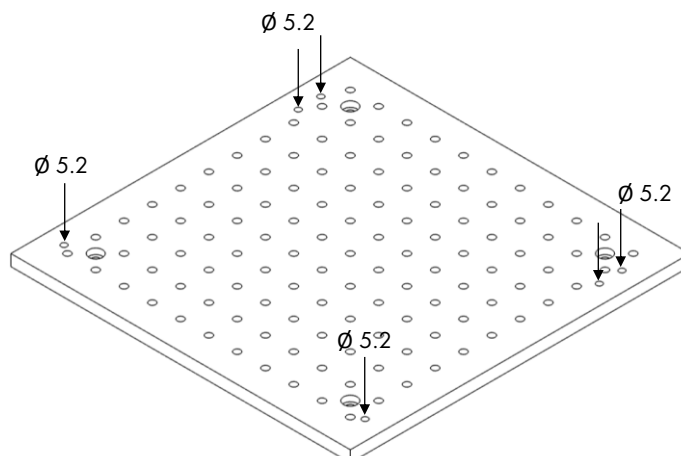
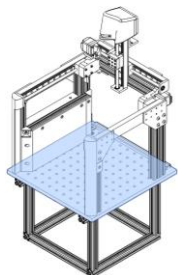
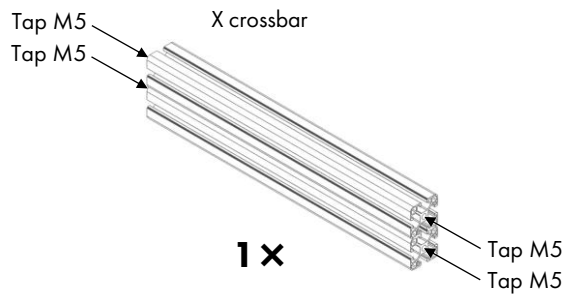
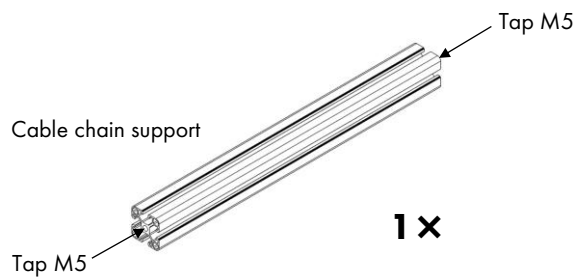
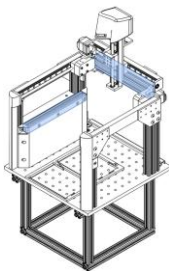
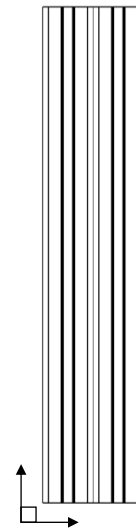
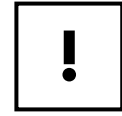
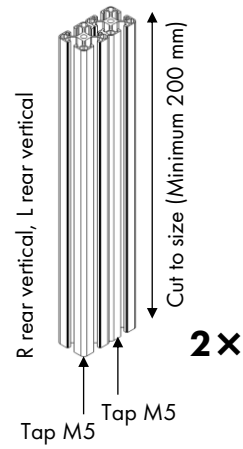
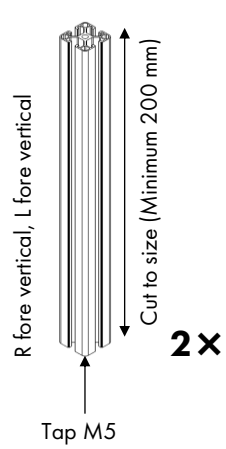
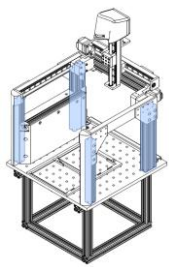


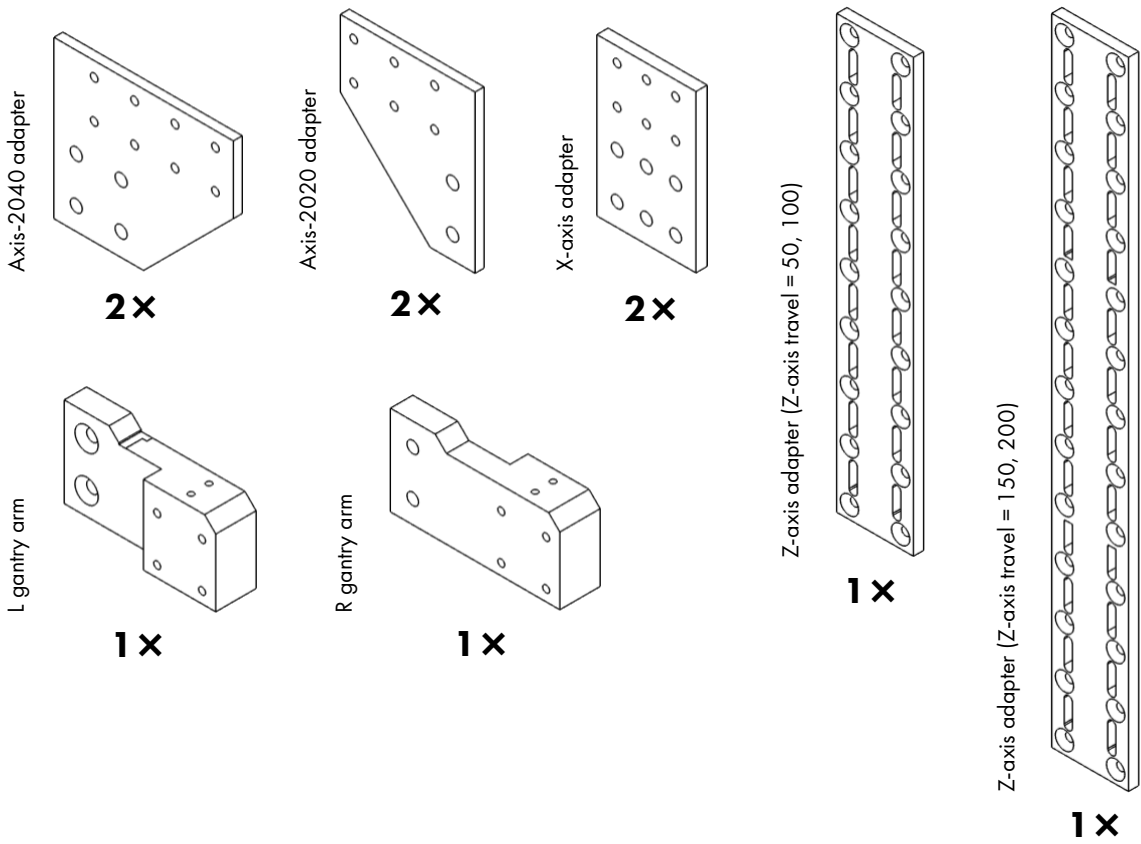
1x ↓

R fore casing

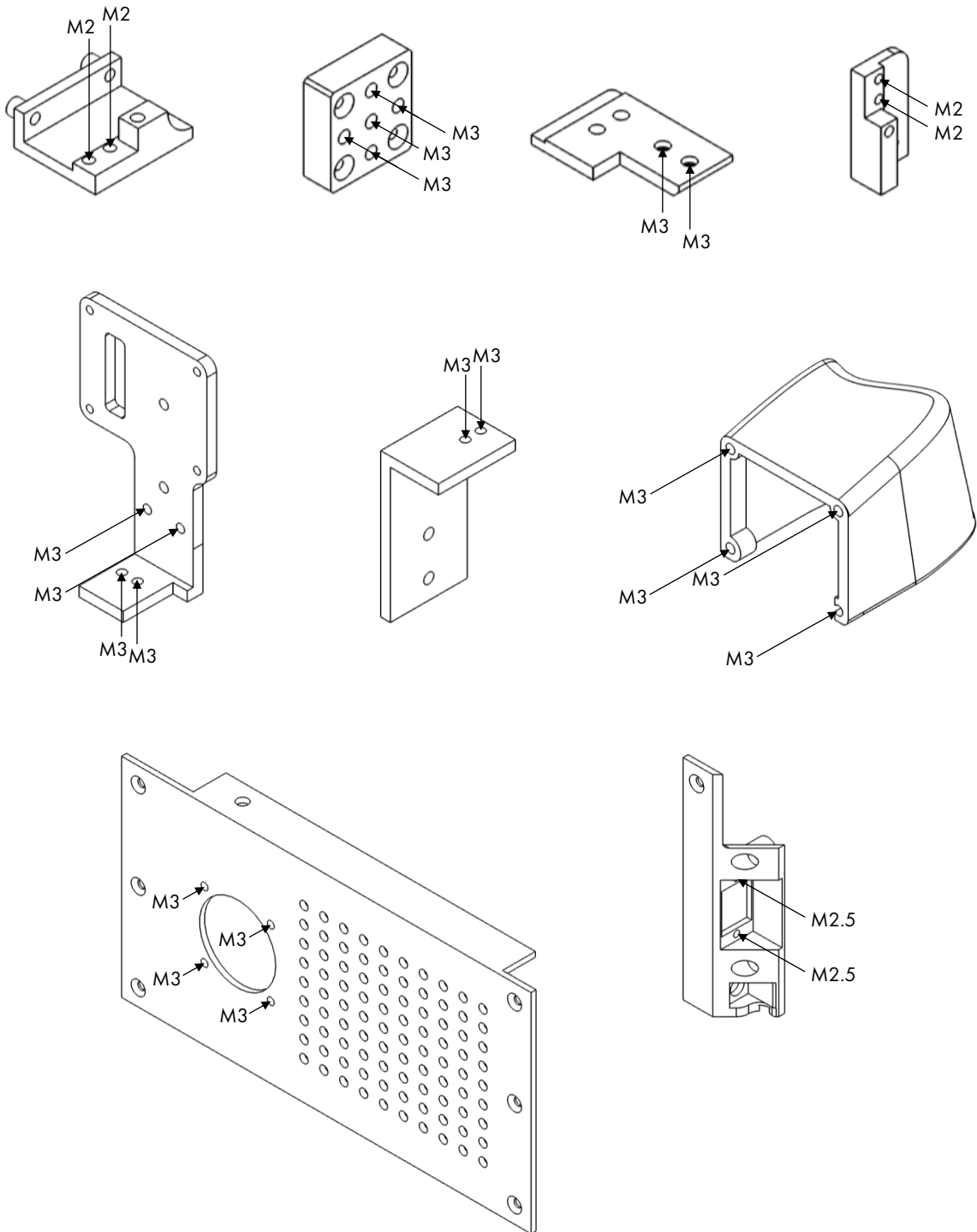


1x ↓



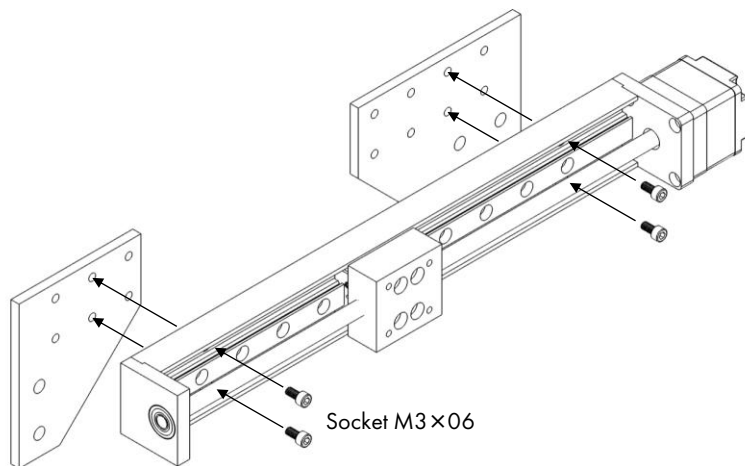


1 HEAT INSERT

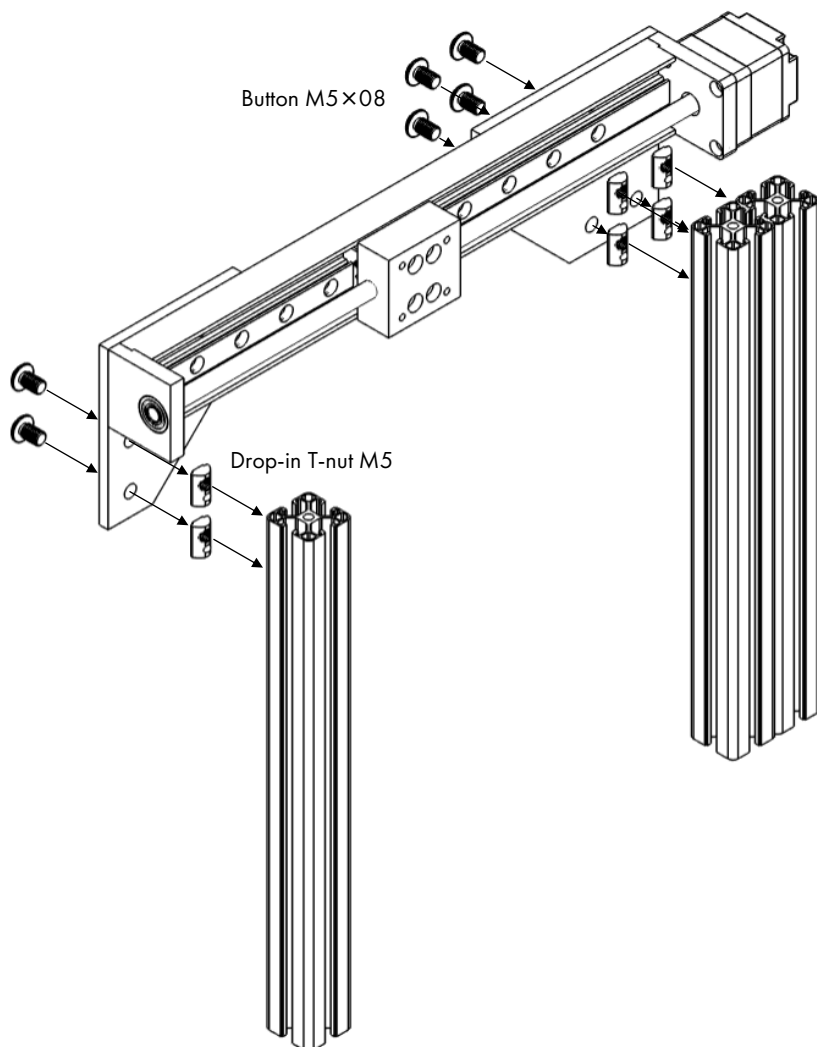


2 ASSEMBLY

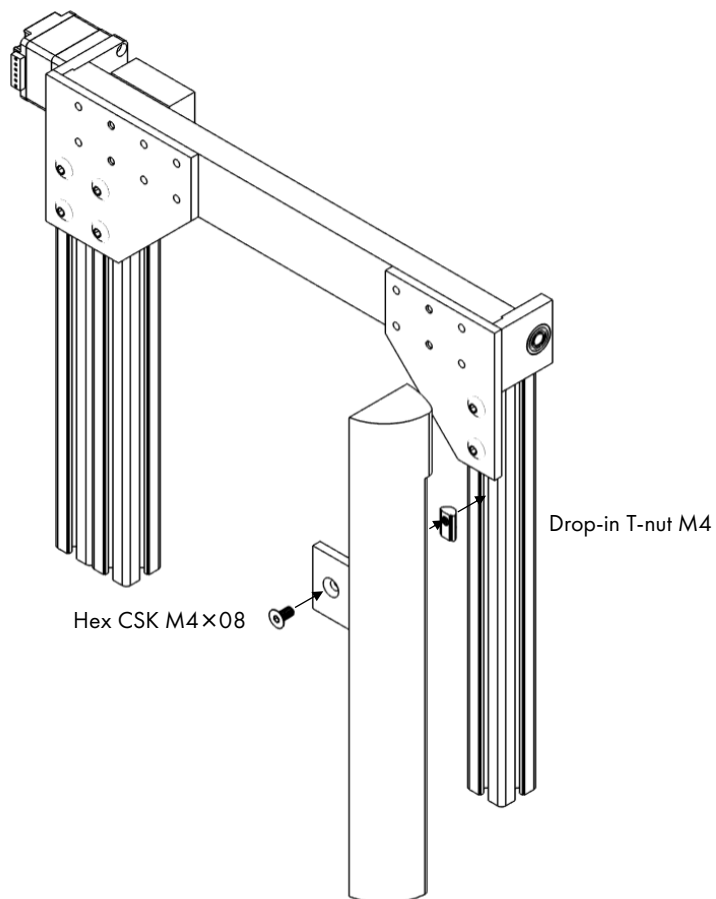
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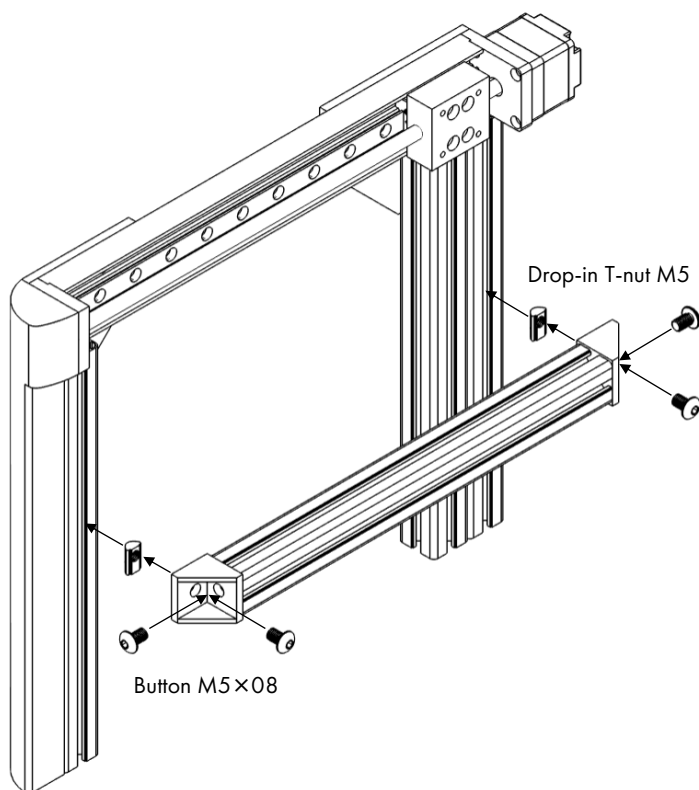
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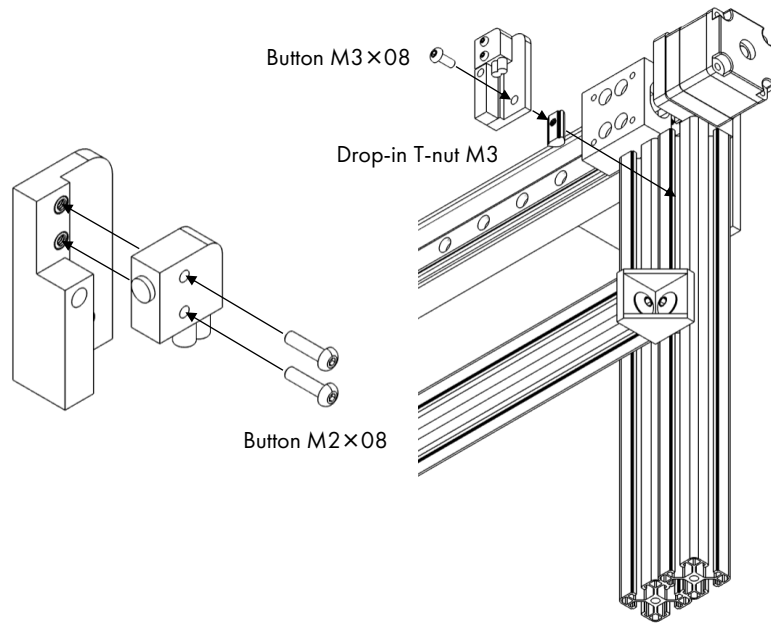
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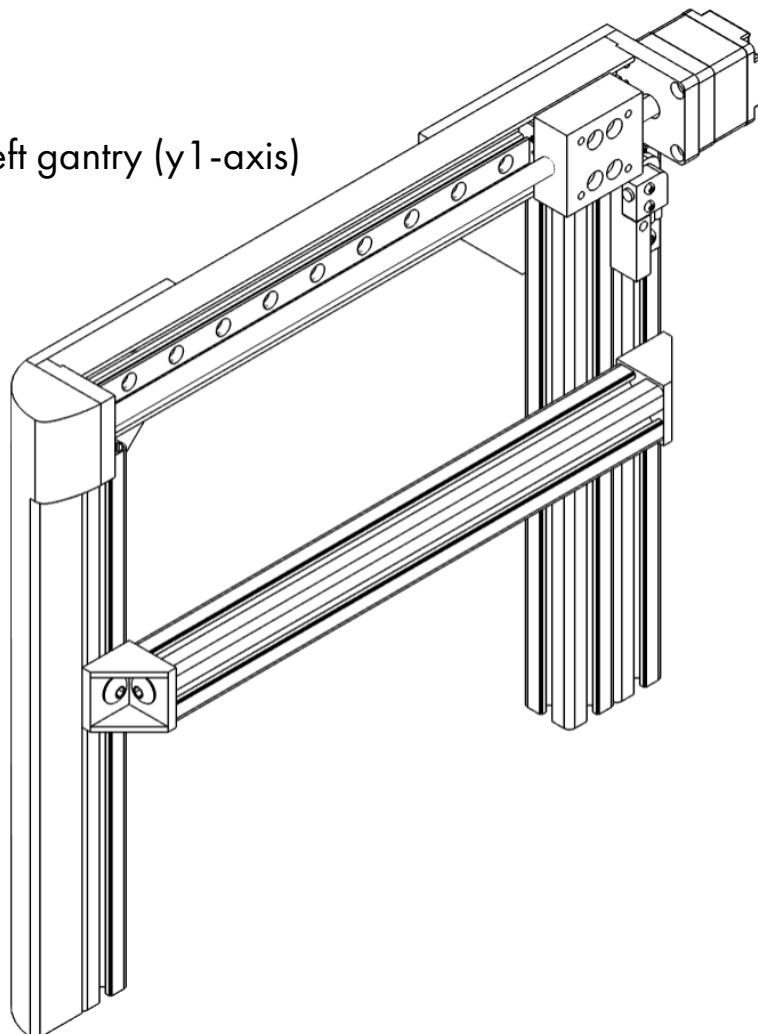
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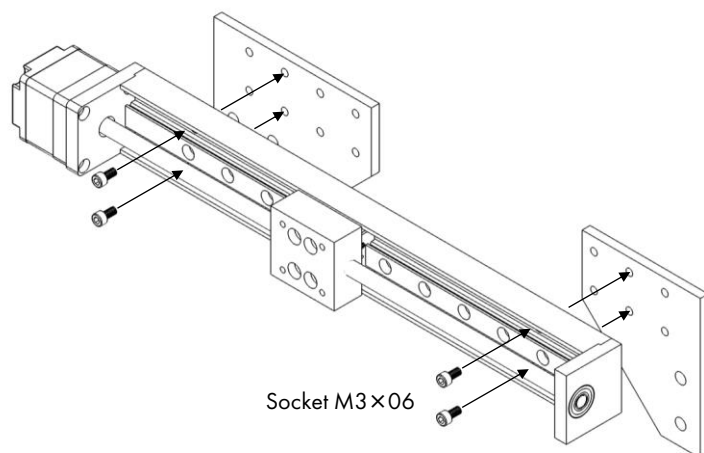
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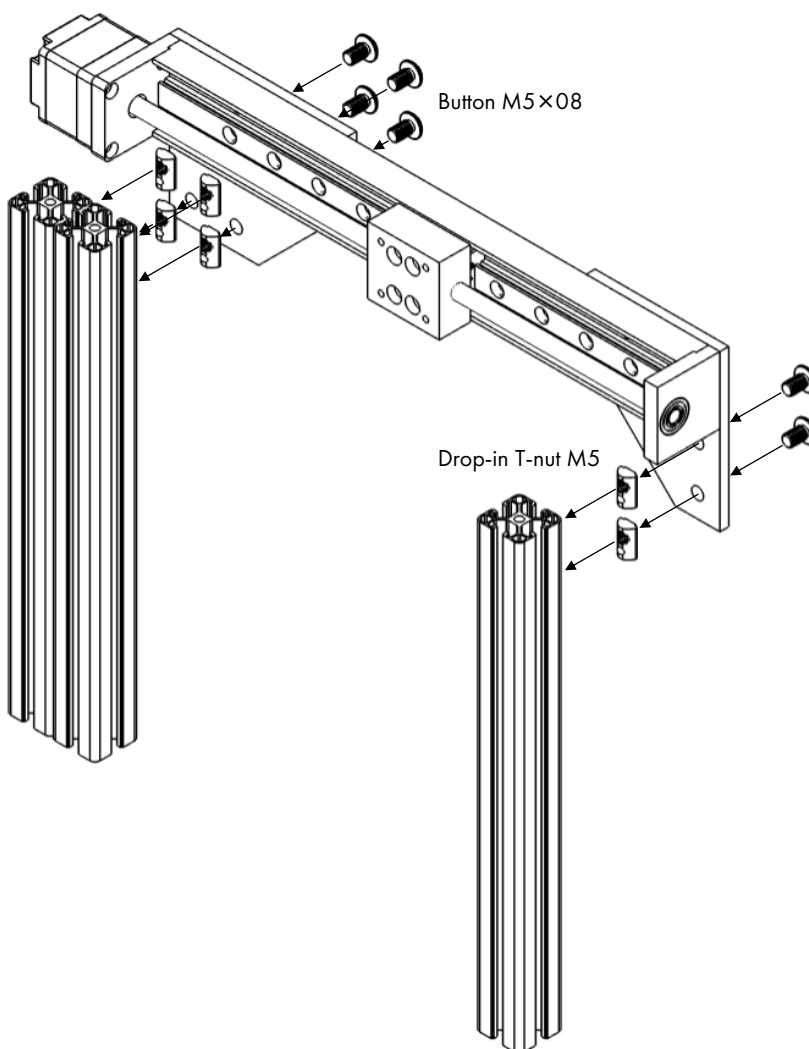
Left gantry (y1-axis)



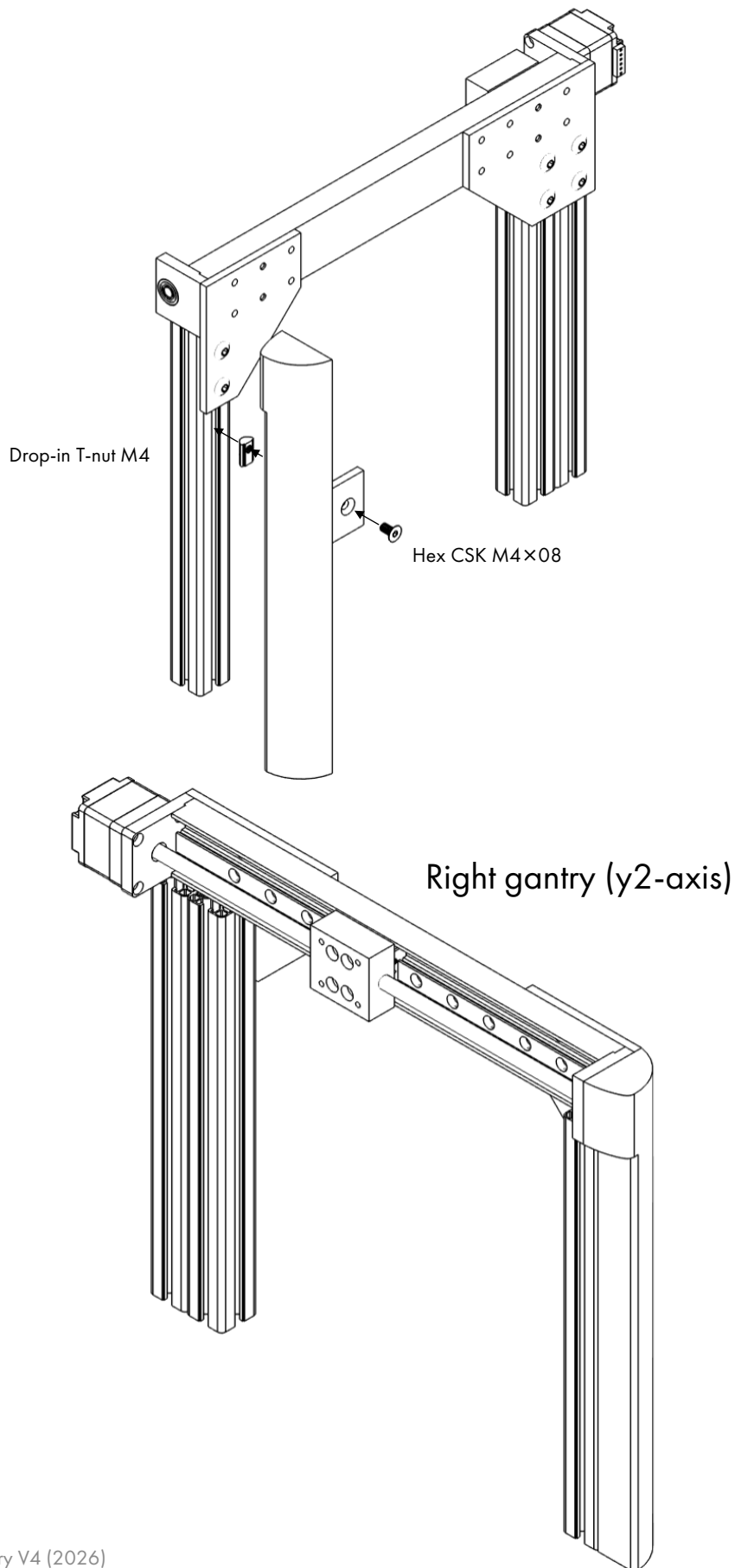
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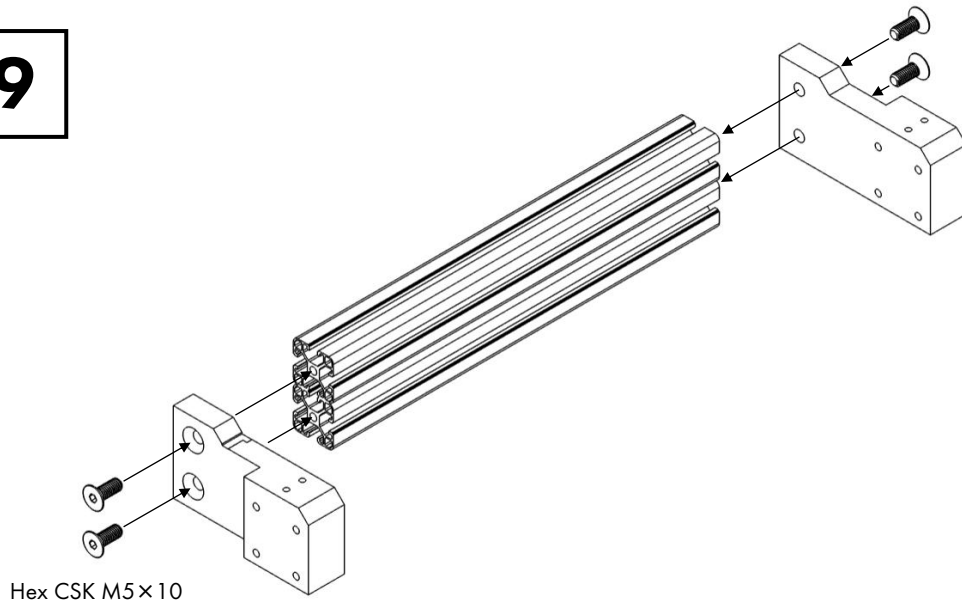
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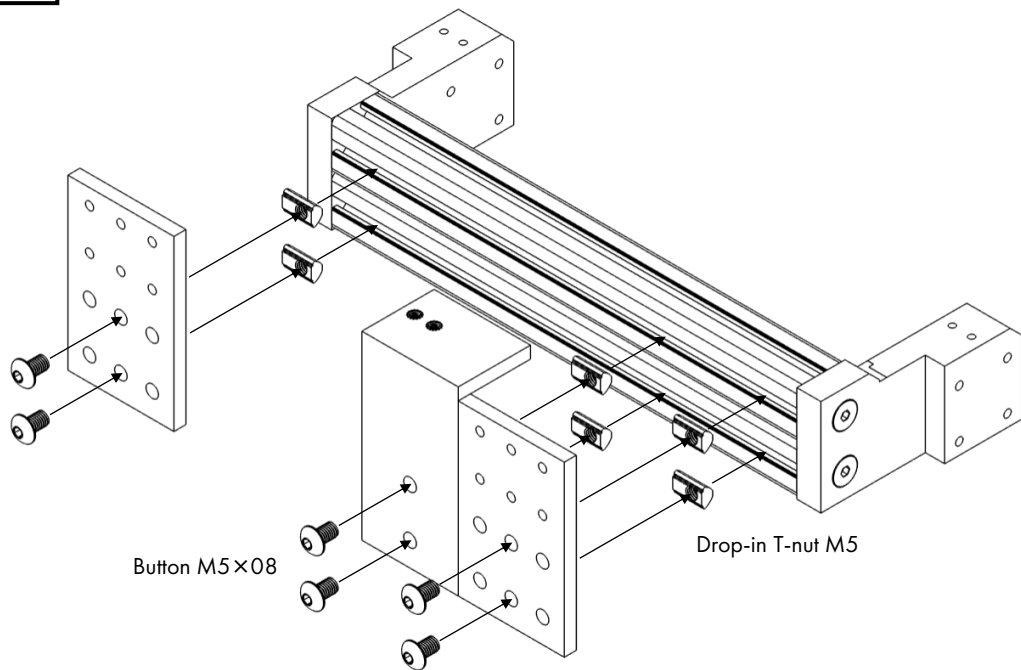
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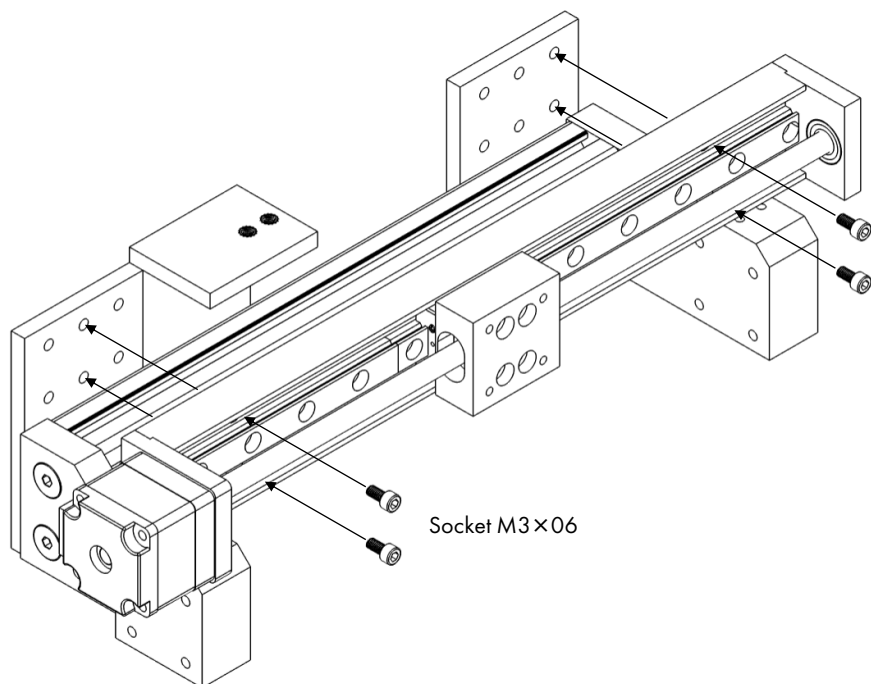
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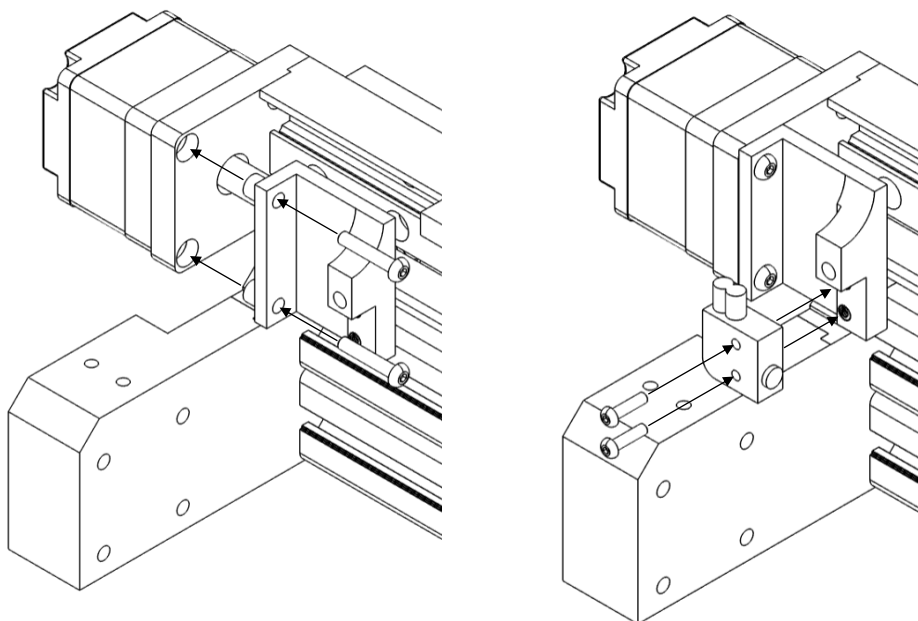
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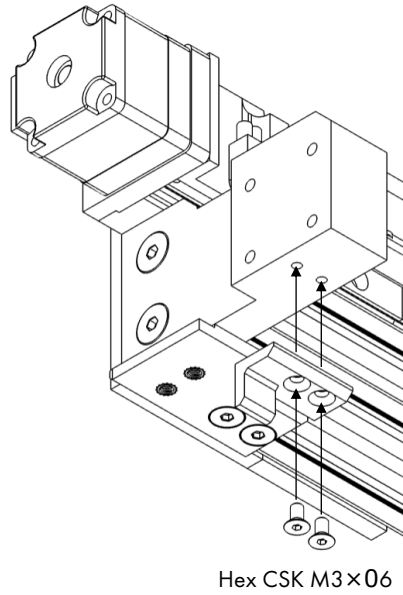
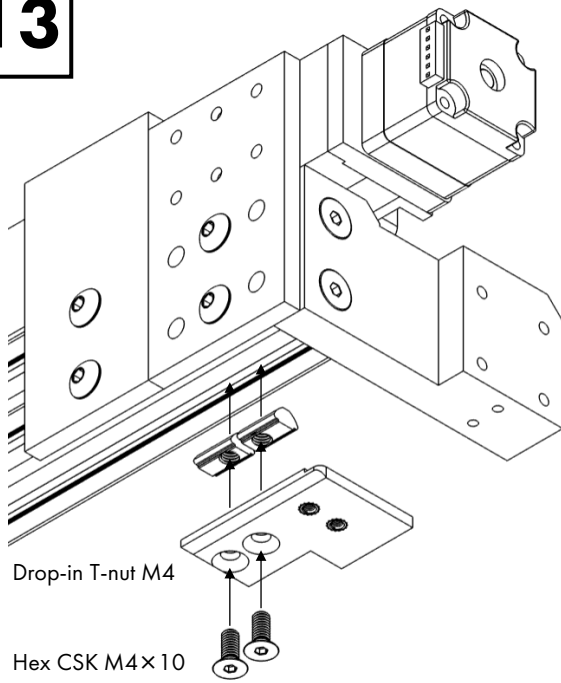
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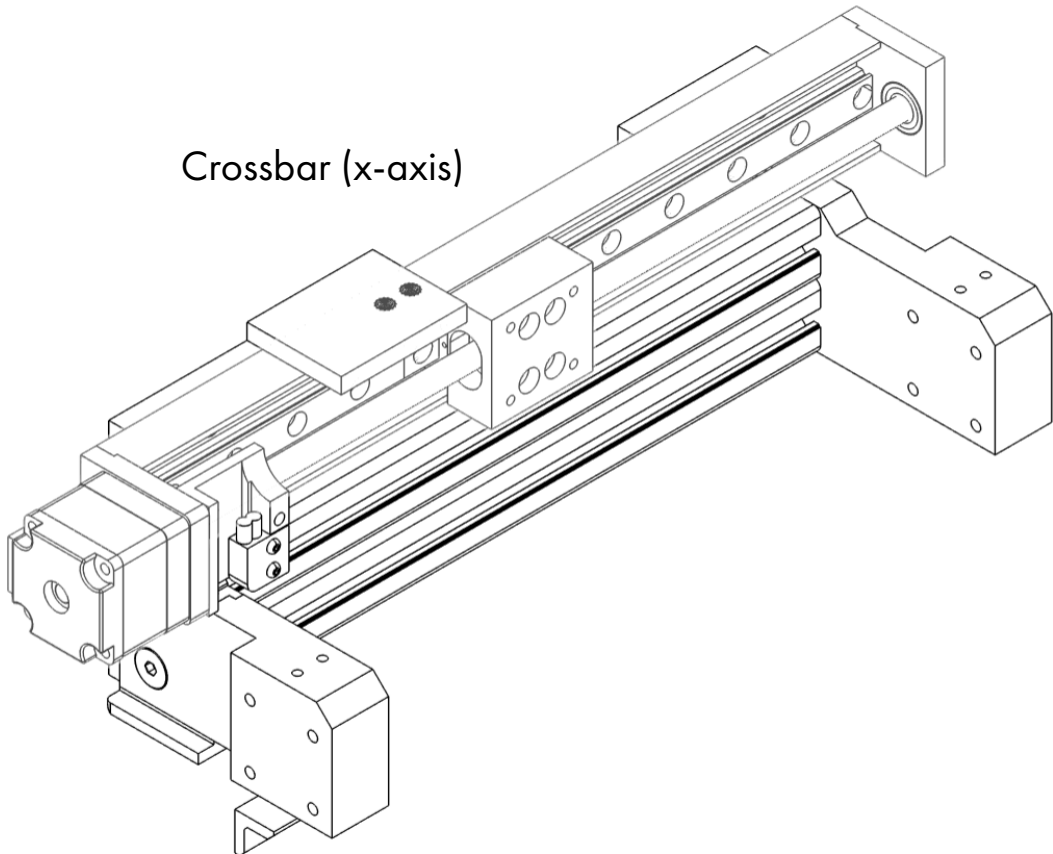
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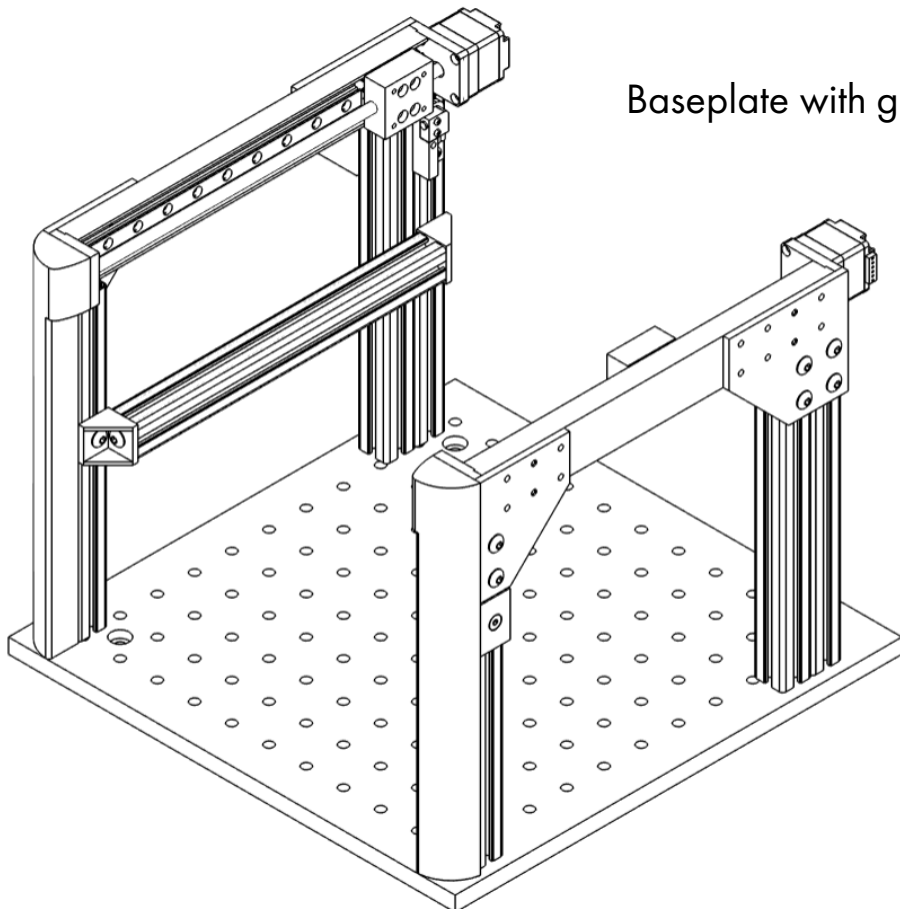
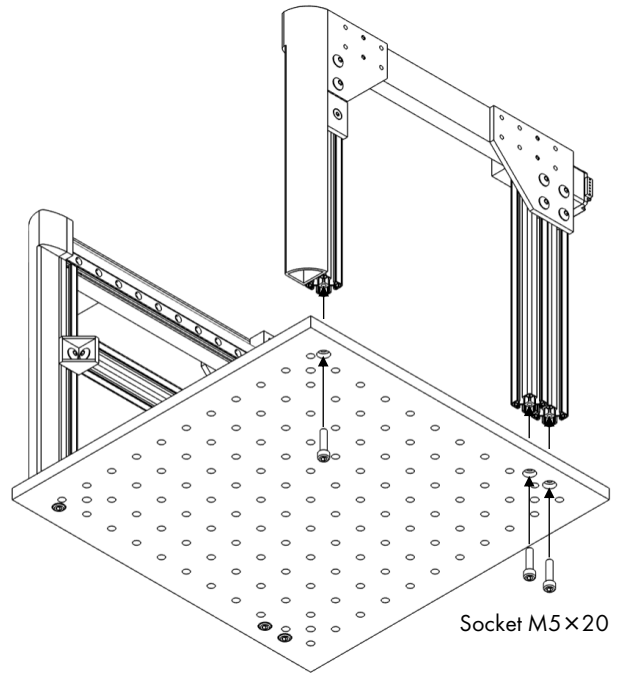
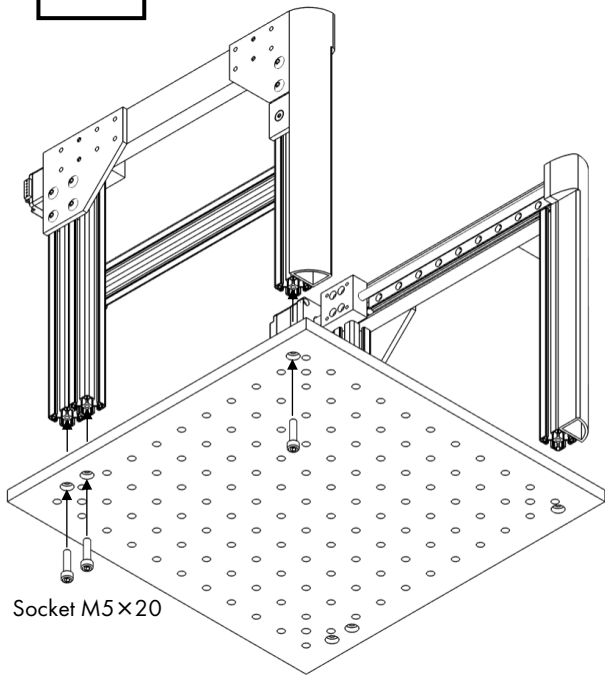


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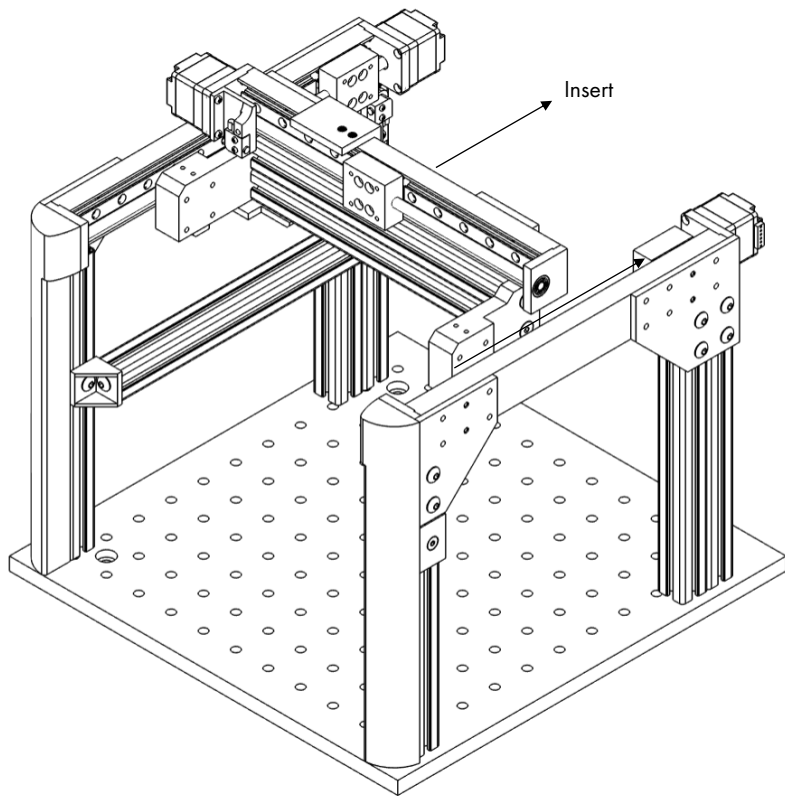
Crossbar (x-axis)



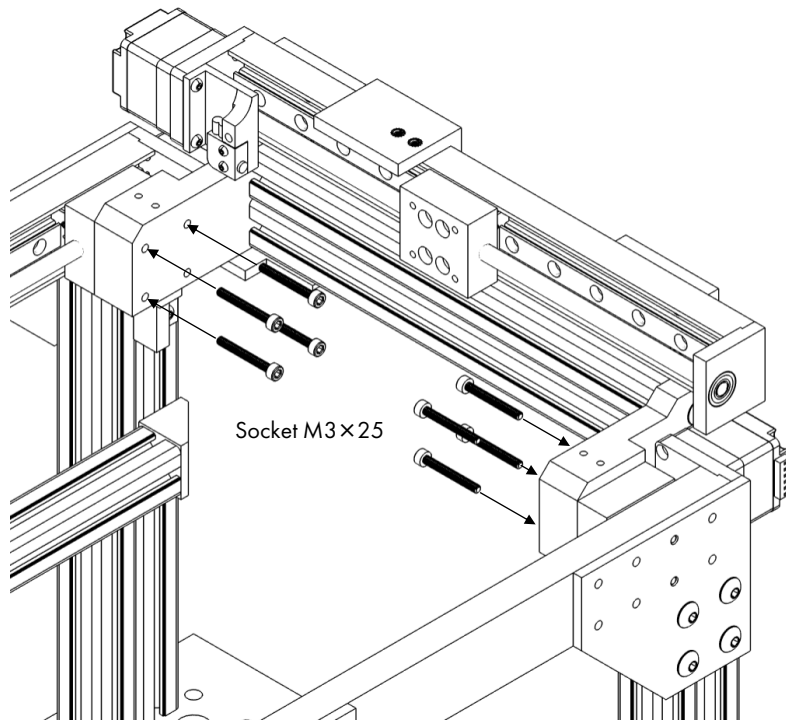


Baseplate with gantries

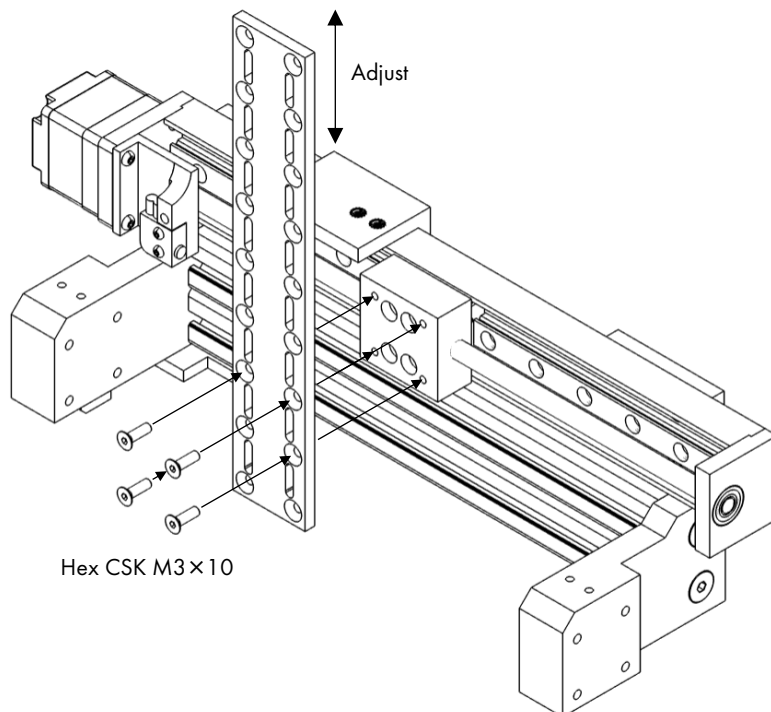
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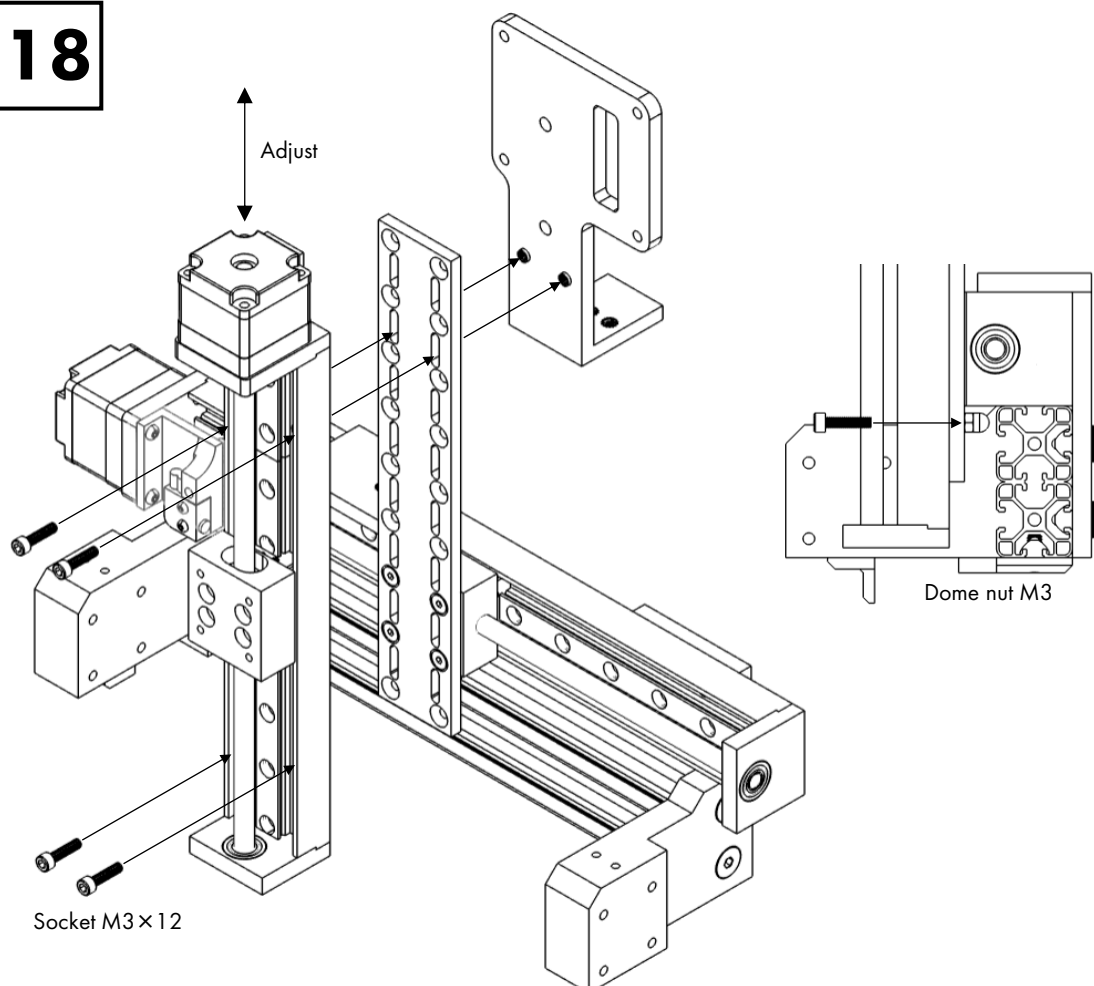
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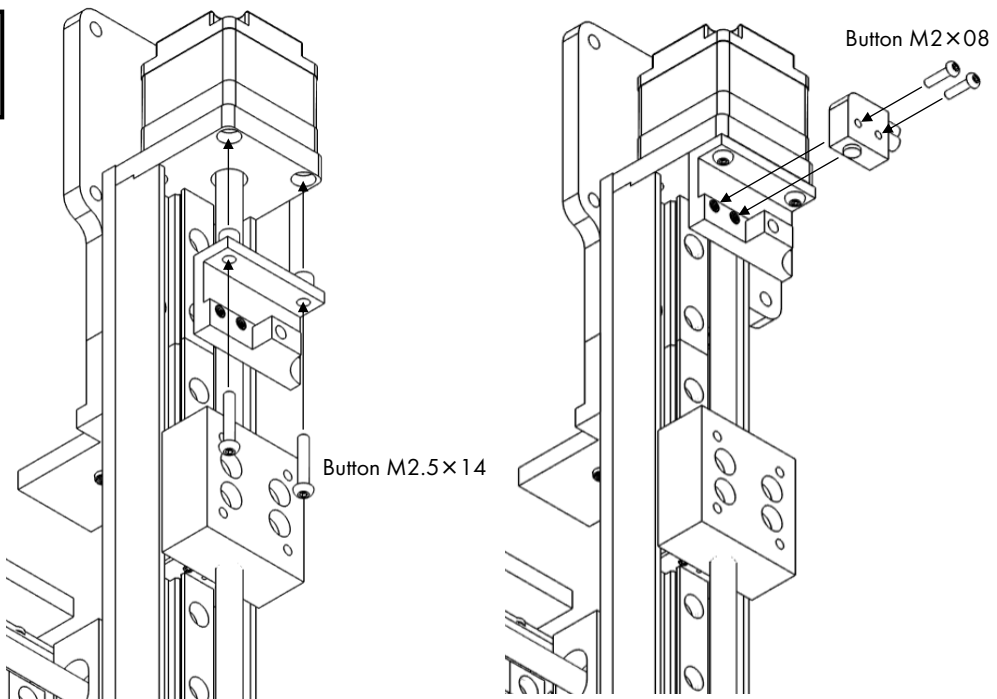
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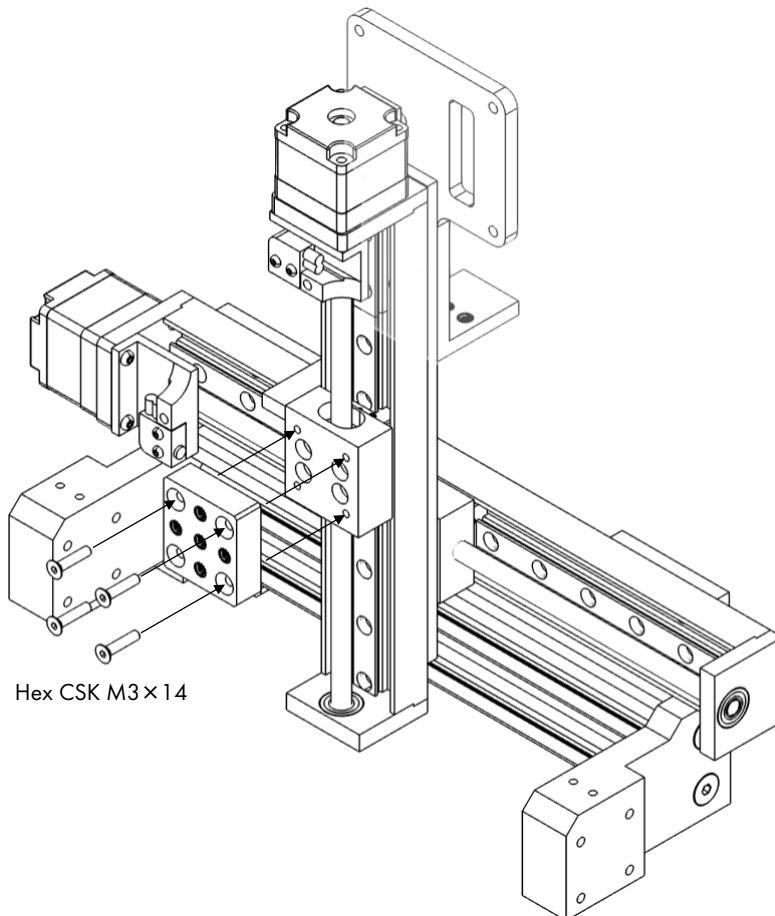
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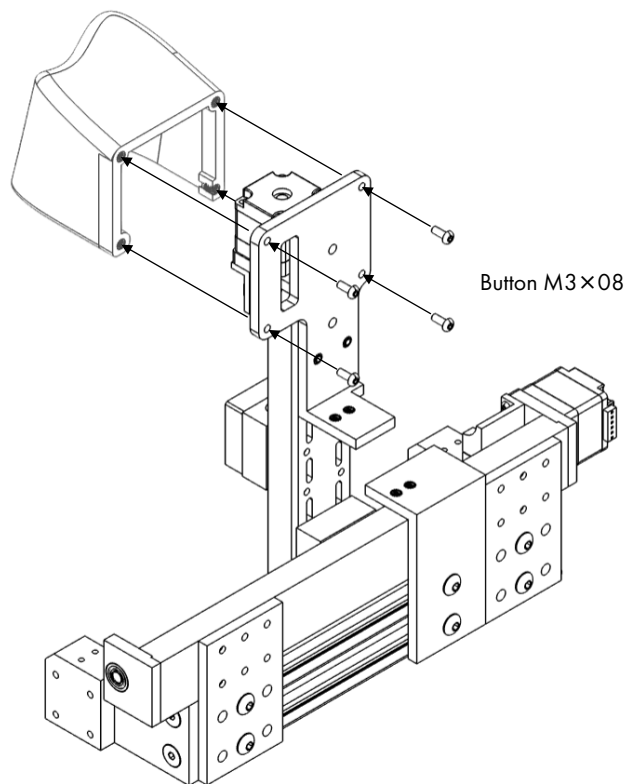
19



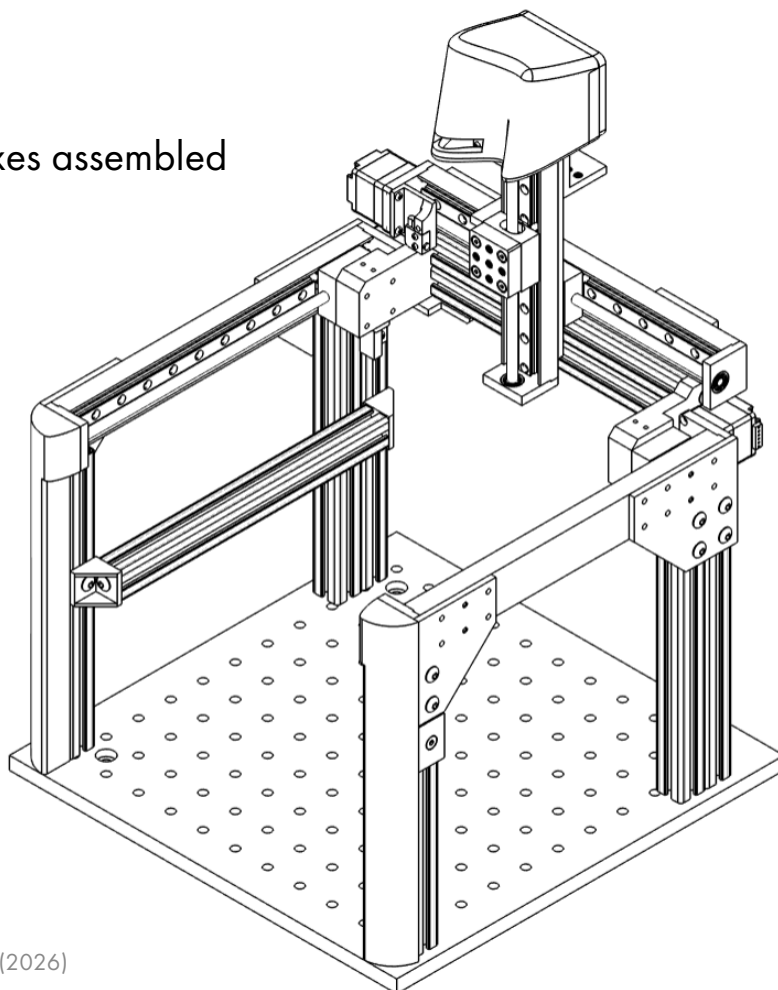
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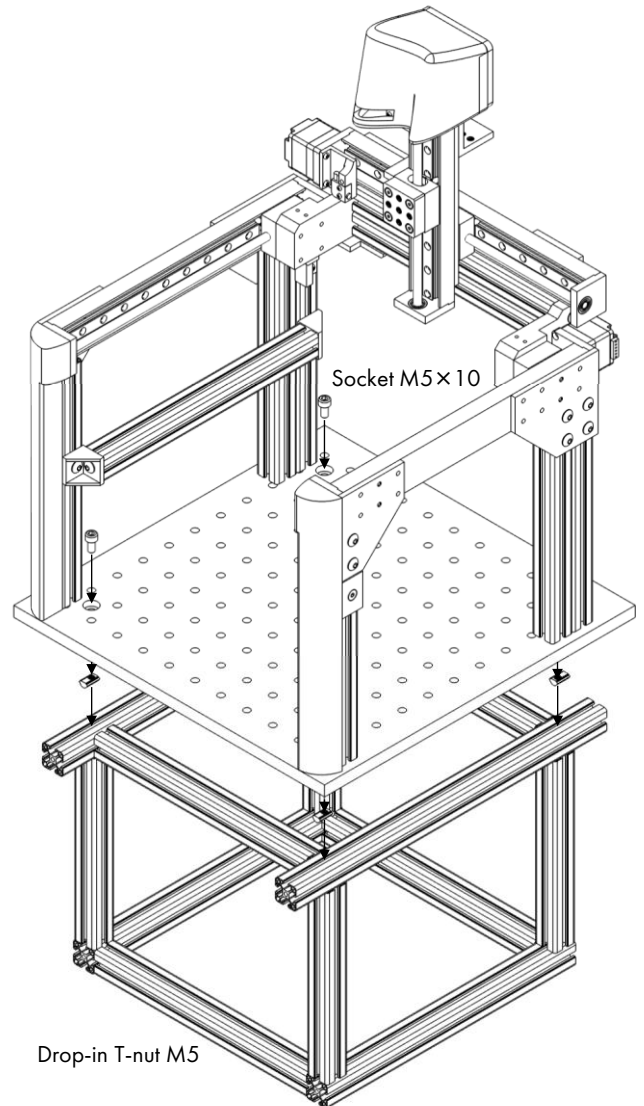
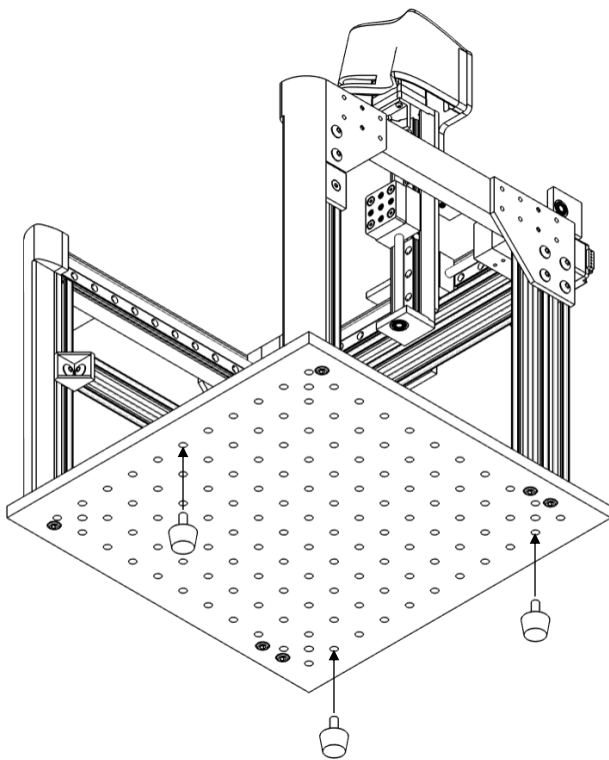
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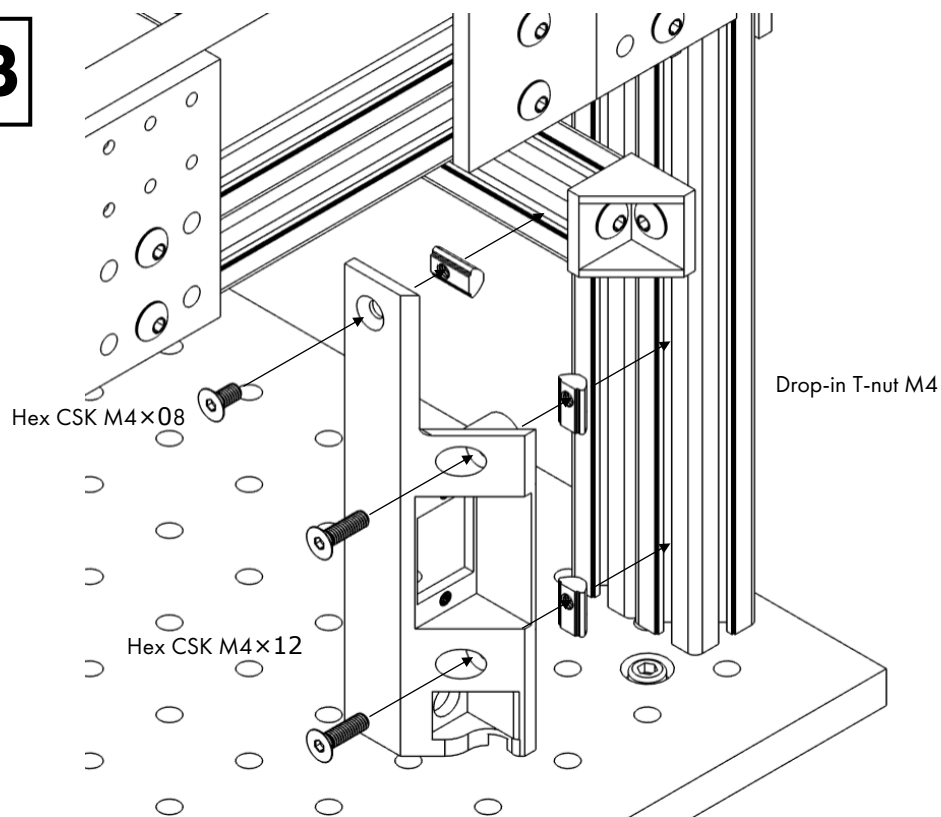
All axes assembled



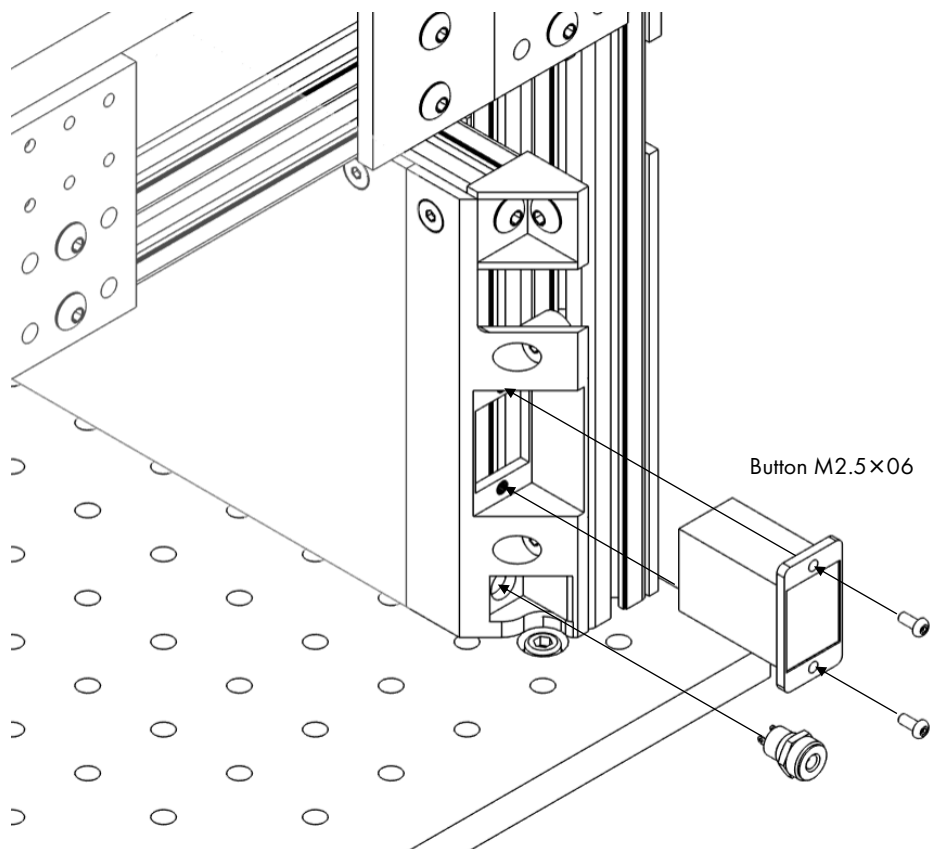
Refer to Utility frame.step
*Customize utility frame per
your mounting setup



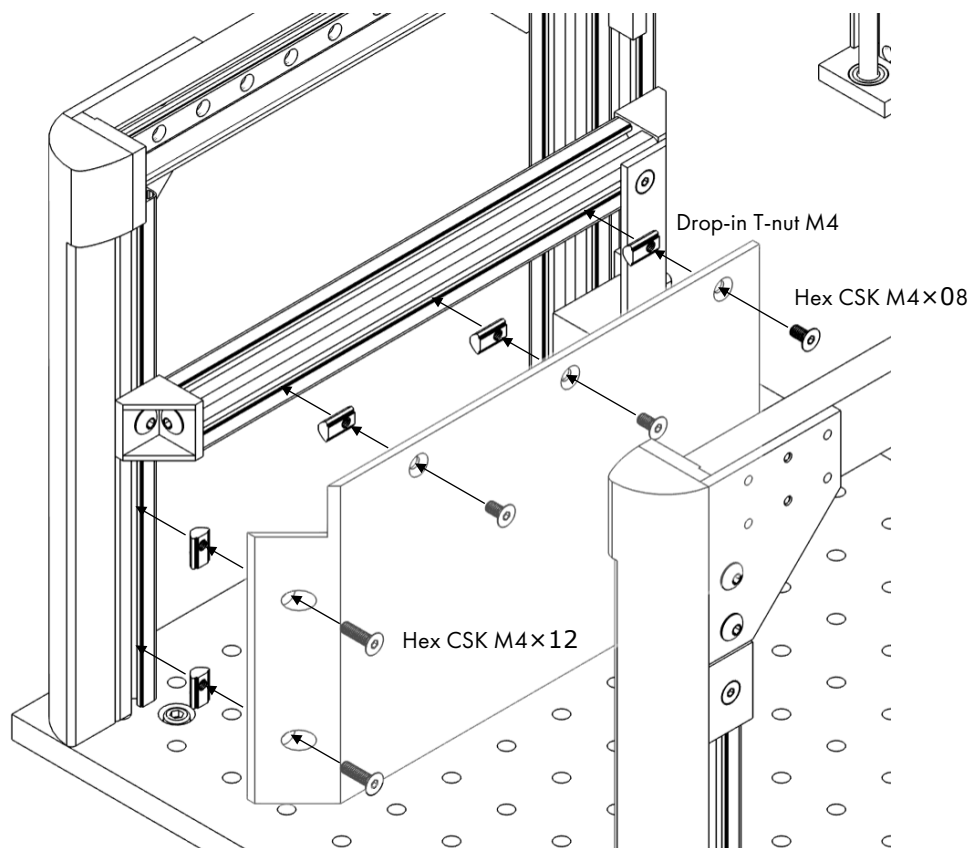
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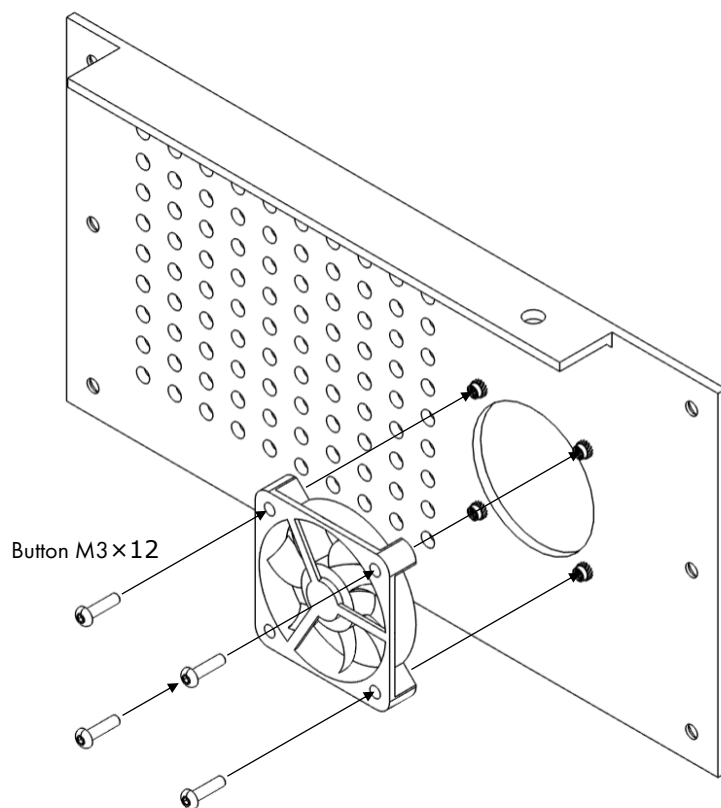
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25



26



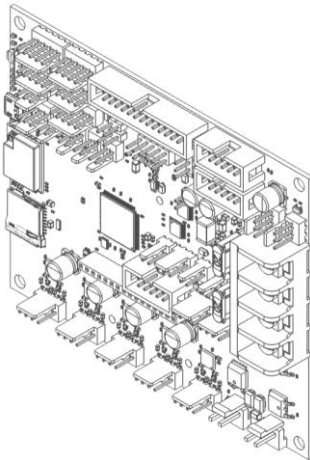
27



Refer to microcontroller and wiring manual

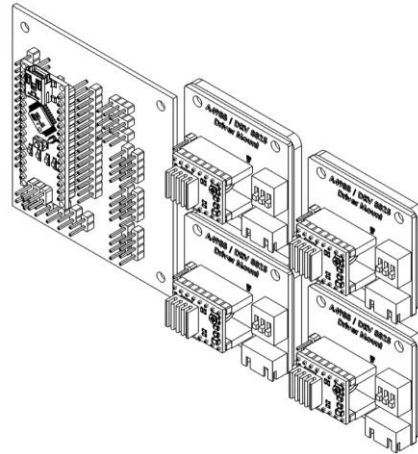
a

Duet 3 Mini 5+ Wifi
\$\$

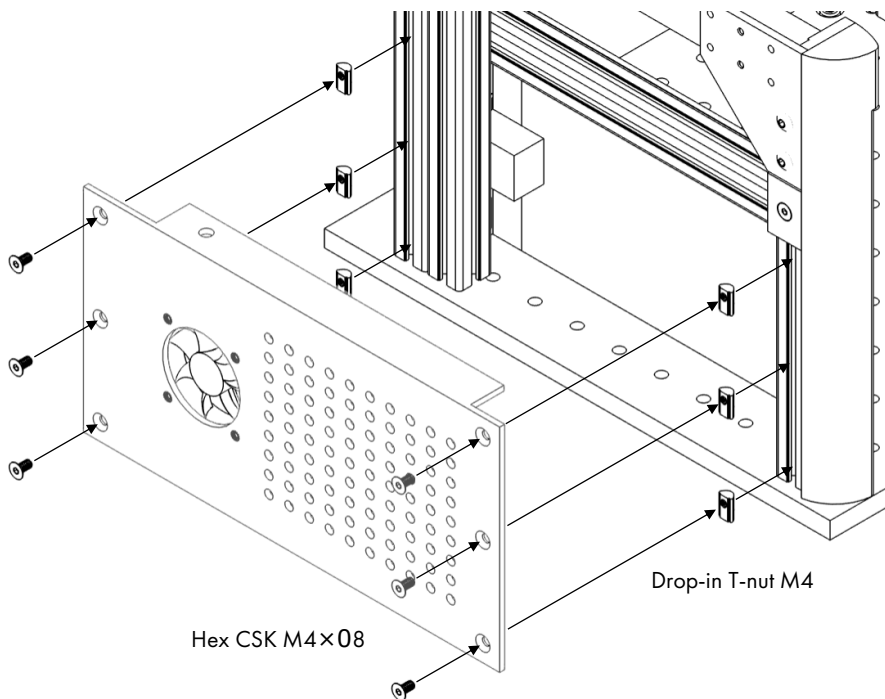


b

Arduino Nano + DRV8825
\$



28



29

Full assembly

